SAHIL VAGHELA

Software Engineer — Real-Time 3D, Systems & XR

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PROFESSIONAL SUMMARY

Software Engineer with 3+ years architecting real-time 3D, spatial computing (Meta Quest 3 / OpenXR), and high-performance game-engine systems across C++, C#, and Python. First Class with Distinction graduate in Computer Engineering (**9.11 CGPA**) with deep expertise delivering Unreal Engine 5 subsystems, procedural CAD-to-3D geometry pipelines, and cross-platform mixed reality applications.

TECHNICAL SKILLS

Programming:	C++, C#, Python, JavaScript / TypeScript, C, SQL
Engines & 3D Systems:	Unreal Engine 5 (C++, Subsystems, Blueprints, Lumen), Unity (C#, URP, Shader Graph), Three.js, WebGL, OpenXR
Software Engineering:	Procedural Mesh Generation, Subsystem Architecture, Data-Driven Design, CAD (DXF/JSON) Parsing, Memory Optimization, REST APIs, Git/GitHub
XR, CV & AI:	Meta Quest 3, Mixed Reality (Passthrough), MediaPipe (Pose Tracking), OpenCV, Vuforia, YOLO11n ONNX, ARCore, Agora Video SDK, WebSockets / UDP
Tools & Platforms:	Visual Studio, Rider, CMake, Blender, Figma, Windows, Linux, Agentic AI Workflows

PROFESSIONAL EXPERIENCE & KEY PROJECTS

Invisible Fiction Gujarat, India
Software & Engine Developer (Promoted: AI Intern → Jr. Unity Developer → Software & Engine Developer) `Feb 2023 – Present`

Fiction Mech (Meta Quest 3 MR & PC Studio — Sole Developer) `Unity • C# • OpenXR • Quest 3 • Agora SDK`

- Architected dual-runtime industrial digital twin platform: PC Authoring Studio in Unity C# (3D CAD ingestion, PBR materials, exploded assembly slider, PDF/MP4 metadata binding, cloud publishing) paired with standalone Meta Quest 3 MR runtime (OpenXR room-scale passthrough, 6DoF hand tracking, cyan outline highlights).
- Authored real-time 3-axis RGB GPU clipping plane shader for mechanical cross-sectioning and integrated Agora SDK telemetry synchronization (<100ms latency) between Quest 3 and Android mobile companion, achieving target 72+ FPS performance on mobile VR hardware.

Radicon Powerbuild AR/MR (Industrial Machinery Showroom — Sole Developer) `Unity • C# • Vuforia • Quest 3 • Android`

- Engineered industrial 3D product showcase on Android and Meta Quest 3, pairing Vuforia printed brochure image-target tracking with markerless ground-plane floor tracking for full-scale factory showroom placement.
- Implemented orbiting 3D product carousel with hero focus, real-time mechanical output shaft swapping on active Series-M model, and in-world curved radial HUD for technical catalogs; optimized mobile fill-rate via CAD decimation and material atlasing.

Fiction ArchViz (Multiuser CAD-to-VR Engine — UE5 C++ & Quest VR) `Unreal Engine 5 • C++ • LibTess2 • DXF/JSON • WebSocket`

- Architected automated CAD-to-3D pipeline parsing multi-layer DXF/JSON blueprints to procedurally generate 3D walls, windows, and roofs via LibTess2 polygon triangulation and gap resolution, eliminating manual 3D modeling turnaround.
- Built 5 predefined showcase rooms with procedural wall tile cutting, 4-way offset stepping, grout customization, DynamicRectLight auto-ceiling illumination, and bidirectional PC ↔ Quest WebSocket multiplayer synchronization via custom Node.js server.

Mimadim Home Studio (High-Fidelity PC Customization — UE5 C++) `Unreal Engine 5 • C++ • Data Assets • Subsystems • Lumen`

- Engineered interior showroom for high-end PC under photorealistic Lumen PBR lighting, creating modular Unreal Data Assets and Master Registry mapping room categories to available beds, parquet flooring, wallpapers, and curtains.
- Developed centralized GameInstance Subsystem with matrix-safe transform preservation and asynchronous asset preloading, eliminating positional drift and memory leaks during runtime actor swaps.

Computer Vision, Enterprise AR & Studio Games `Unity C# • MediaPipe • Python • Vuforia • UDP Sockets`

- Streamed 33-point real-time MediaPipe skeletal landmarks over UDP sockets into Unity at 30+ FPS for controller-free motion games (*Alien Runner*, *Football Cup*), establishing the network IPC bridge productized from initial intern prototype.
- Stabilized level-progression state machines, physics, and UI across commercial titles (*Turbo Trail, Cupping Muffins*), shipped casual arcade game *Catch the Object* as sole developer, and deployed QR-triggered AR model visualization on Microsoft HoloLens with Vuforia and ARCore.

EDUCATION & ACADEMIC ACHIEVEMENTS

Bachelor of Engineering (B.E.) in Computer Engineering `Graduated: 2023`
Ipcowala Institute of Engineering and Technology — Gujarat, India CGPA: 9.11 / 10.0
Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Computer Graphics, Database Systems, Operating Systems, Computer Networks. • **Languages:** English (Professional), Hindi, Gujarati.